SQL Notes for TCO RAG Testing

Synthetic educational notes created specifically for testing the TCO document loader, chunking, retrieval, and grounded assessment workflow.

1. What is SQL?

SQL (Structured Query Language) is a language used to communicate with and work with relational databases. It can be used to retrieve, insert, update, and delete data, and to create or modify database structures.

Relational Database

A relational database stores information in tables. A table contains rows and columns. A row represents a record, while a column represents an attribute of that record.

Important terms

Term	Meaning
Table	Collection of related data organized in rows and columns.
Row	A single record in a table.
Column	An attribute or field in a table.
Primary Key	A column or set of columns used to uniquely identify a row.
Foreign Key	A column that refers to a key in another table.

2. SELECT Statement

The SELECT statement is used to retrieve data from one or more tables.

Example:

```
SELECT * FROM Employees;
```

The asterisk (*) means that all columns are requested.

To select particular columns:

```
SELECT Name, Salary FROM Employees;
```

3. WHERE Clause

The WHERE clause filters rows according to a condition.

Example:

```
SELECT * FROM Employees WHERE Department = 'Sales';
```

Multiple conditions can be combined using AND or OR.

```
SELECT * FROM Employees WHERE Department = 'Sales' AND Salary > 50000;
```

4. INSERT, UPDATE, and DELETE

INSERT

INSERT is used to add new rows to a table.

```
INSERT INTO Students (Name, Age) VALUES ('Rohan', 20);
```

UPDATE

UPDATE changes existing data in a table. A WHERE condition is commonly used to limit which rows are changed.

```
UPDATE Students SET Age = 21 WHERE Name = 'Rohan';
```

DELETE

DELETE removes selected rows from a table when a condition is supplied.

```
DELETE FROM Students WHERE Name = 'Rohan';
```

5. DELETE vs TRUNCATE vs DROP

Command	Main purpose
DELETE	Removes rows from a table; a WHERE clause can target specific rows.
TRUNCATE	Removes all rows from a table while keeping the table structure.
DROP	Removes the table structure and its data.

6. ORDER BY

ORDER BY sorts query results by one or more columns.

```
SELECT Name, Salary FROM Employees ORDER BY Salary DESC;
```

DESC sorts from higher to lower values. ASC sorts from lower to higher values.

7. GROUP BY and Aggregate Functions

GROUP BY places rows with the same values into groups. It is commonly used with aggregate functions.

Common aggregate functions include COUNT(), SUM(), AVG(), MAX(), and MIN().

```
SELECT Department, COUNT(*) FROM Employees GROUP BY Department;
```

8. SQL JOINs

A JOIN combines rows from two or more tables based on a related column.

JOIN	Meaning
INNER JOIN	Returns rows that have matching values in both joined tables.
LEFT JOIN	Returns all rows from the left table and matching rows from the right table; unmatched right-side values can b

INNER JOIN example

```
SELECT Employees.Name, Departments.DepartmentName FROM Employees INNER JOIN Departments
ON Employees.DepartmentID = Departments.DepartmentID;
```

LEFT JOIN example

```
SELECT Employees.Name, Departments.DepartmentName FROM Employees LEFT JOIN Departments ON
Employees.DepartmentID = Departments.DepartmentID;
```

9. Database Structure Commands

DDL (Data Definition Language) is used to define or change database structures. Common commands include CREATE, ALTER, and DROP.

```
CREATE TABLE Students (StudentID INT, Name VARCHAR(100), Age INT);

ALTER TABLE Students ADD Email VARCHAR(150);
```

10. Normalization Basics

Normalization is a database design approach used to organize data and reduce unnecessary redundancy. Common normal forms include 1NF, 2NF, and 3NF.

1NF: Each field should contain a single, atomic value rather than repeating groups.

2NF: A table should be in 1NF and non-key attributes should depend on the whole candidate key in the relevant composite-key case.

3NF: A table should be in 2NF and non-key attributes should not depend transitively on another non-key attribute.

11. Reference Questions for RAG Testing

1. What does SQL stand for and what is its primary purpose?
2. Which SQL statement is used to retrieve data?
3. What is the purpose of the WHERE clause?
4. Explain the difference between DELETE, TRUNCATE, and DROP.
5. Write a query to retrieve employees from the Sales department.
6. What is a SQL JOIN?
7. What is the difference between INNER JOIN and LEFT JOIN?
8. What is the purpose of GROUP BY?
9. Name common aggregate functions used with GROUP BY.
10. What is normalization and why is it used?

12. RAG Testing Topics

Useful query phrases for testing retrieval include: SQL SELECT WHERE clause; DELETE TRUNCATE DROP; SQL JOINs; INNER JOIN vs LEFT JOIN; GROUP BY aggregate functions; database normalization 1NF 2NF 3NF.

This PDF is intentionally self-contained and synthetic so that the TCO RAG pipeline can be tested without relying on an external source.